

SupplyTwinAI: An Autonomous Multi-Agent Supply Chain Digital Twin for Intelligent Decision Support

Abstract:

Supply chains are increasingly vulnerable to disruptions caused by supplier failures, transportation delays, fluctuating demand, adverse weather conditions, and geopolitical events. Traditional supply chain management systems often lack real-time visibility, intelligent decision-making, and proactive mitigation capabilities. The IEEE paper “*Digital Twins of Supply Chains: A Systems Approach*” proposes a digital twin architecture for monitoring and synchronizing supply chain operations. However, the framework primarily focuses on digital representation and data integration, with limited support for autonomous reasoning and adaptive response mechanisms.

To address these limitations, this project proposes **SupplyTwinAI**, an AI-driven autonomous supply chain digital twin framework. The proposed system integrates Knowledge Graphs, GraphRAG, LangGraph-based multi-agent orchestration, persistent disruption memory, and external intelligence sources such as weather forecasts and news feeds. By leveraging Large Language Models, the system continuously monitors supply chain events, analyzes potential disruptions, and recommends mitigation strategies in real time. SupplyTwinAI aims to enhance supply chain resilience, improve operational visibility, and support intelligent decision-making for efficient and adaptive supply chain management.

Keywords:

Supply Chain Digital Twin, Digital Twin Architecture, Knowledge Graph, GraphRAG, LangGraph, Multi-Agent Systems, Supply Chain Risk Management, Autonomous Decision Support, Supply Chain Resilience, Large Language Models (LLMs), Artificial Intelligence.

Team No: 12

Domain of the Project: Artificial Intelligence and Machine Learning

Team Members:

S. No.	Roll No	Student Name	Phone no
1.	23BD1A0566	Pranitha Avula	9959241516
2.	23BD1A0579	Sindhu Sri Thirunagari	9866180360
3.	23BD1A057A	Kavya Tummalapally	8688996407
4.	23BD1A057H	Divya Sree Vavilala	7989305437

Faculty Supervisor

Project Coordinator